

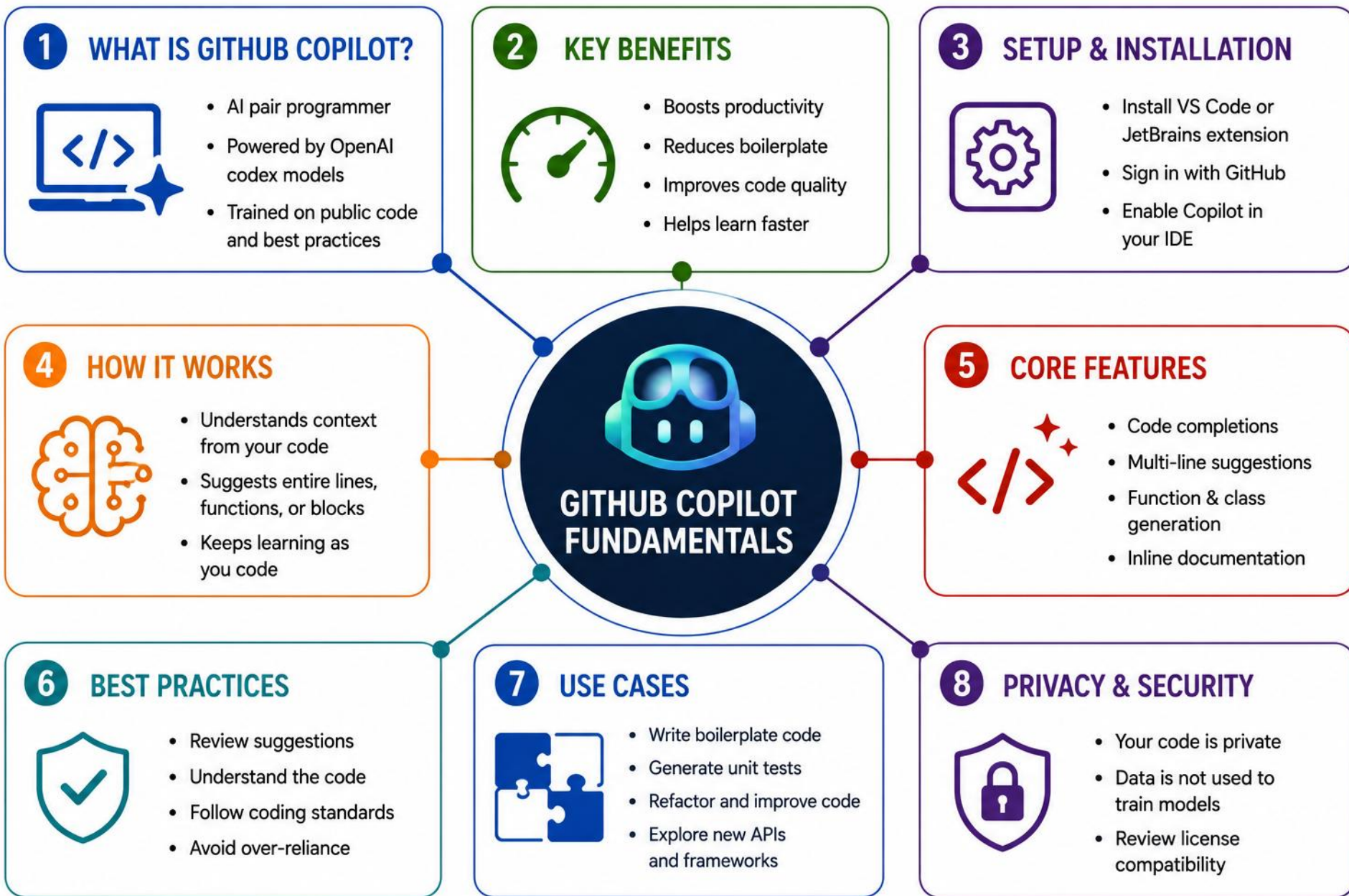
MODULE 10

GitHub Copilot Fundamentals for Java Developers

Use GitHub Copilot to accelerate development, write better code, and improve productivity — responsibly.







MODULE 10 OVERVIEW

AI-assisted development is now part of enterprise software engineering — developers must learn when to trust AI, and when to verify it.

- A developer needs to rapidly scaffold a new customer service. Copilot accelerates development — but every suggestion still needs a human reviewer.

DURATION & LAB



2 Hours Theory

How Copilot works, prompting, generation, review.



45 Min Lab

AI-Assisted Code Generation.



Business Scenario

Scaffold the Northstar CRM customer domain with Copilot.

COPILOT IS

A context-aware AI pair programmer, not an autopilot

YOU DECIDE

Every suggestion is reviewed, tested, and refined by you

THE GOAL

Write better code, faster, without losing engineering judgment

KEY TAKEAWAY AI is a force multiplier — it augments your skills, it doesn't replace your judgment.

LEARNING OBJECTIVES

By the end of this module, you will use GitHub Copilot effectively to write, explain, refactor, and test code.

- 1 Understand what GitHub Copilot is and how it works.
- 2 Set up and configure Copilot in your dev environment.
- 3 Generate context-aware code and complete tasks faster.
- 4 Write clear prompts for better, more accurate suggestions.
- 5 Explain and document code with natural language.
- 6 Refactor and improve code using Copilot.
- 7 Generate tests and handle edge cases.
- 8 Review and validate AI-generated code critically.
- 9 Understand risks, limitations, and ethical use.
- 10 Apply Copilot to real enterprise use cases.

KEY BENEFITS



Boost Productivity



Improve Code Quality

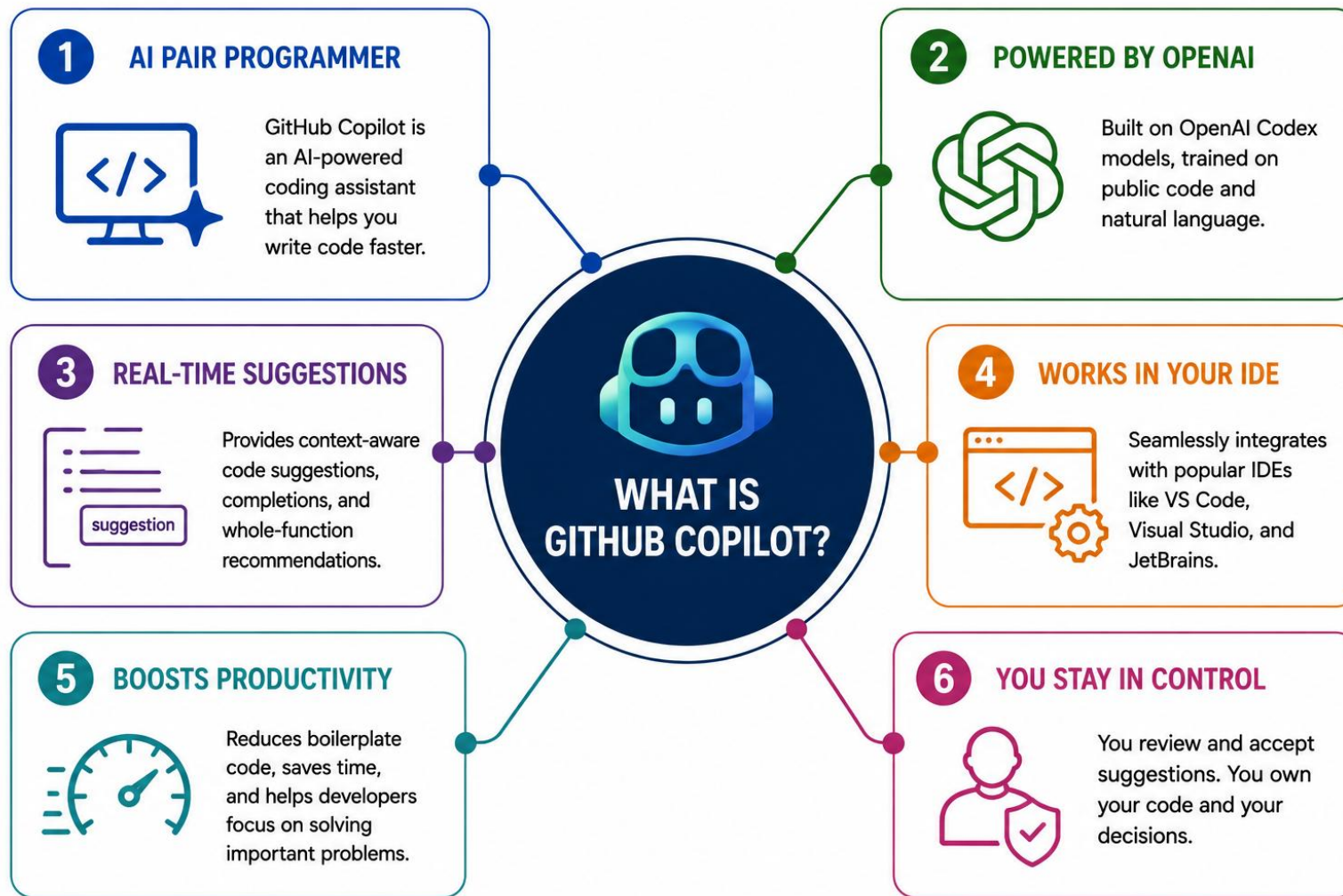


Accelerate Learning



Focus on Impact

KEY TAKEAWAY You will be confident using GitHub Copilot to write better code, move faster, and deliver high-quality software.

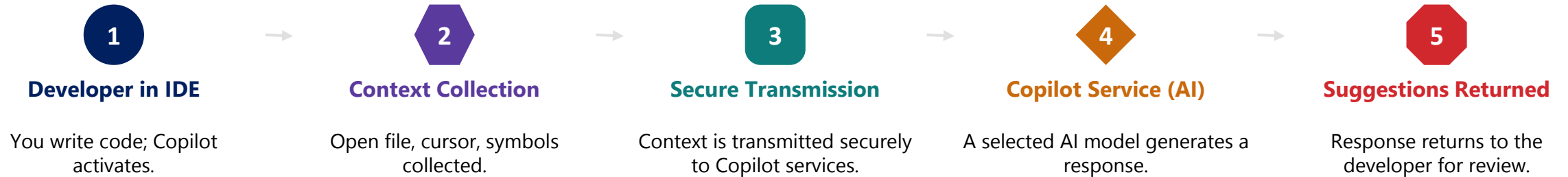


WHAT IS GITHUB COPILOT GitHub Copilot is an AI coding assistant that uses editor and repository context to generate suggestions, explanations, and code-related responses.

It uses large language models selected and operated through GitHub's service, and adapts its suggestions to the context available in your editor, files, repository, instructions, and conversation — not by retraining on your code.

COPILOT ARCHITECTURE OVERVIEW

Copilot integrates into your IDE, securely sends context to GitHub's AI service, and returns intelligent suggestions in real time.



KEY COMPONENTS

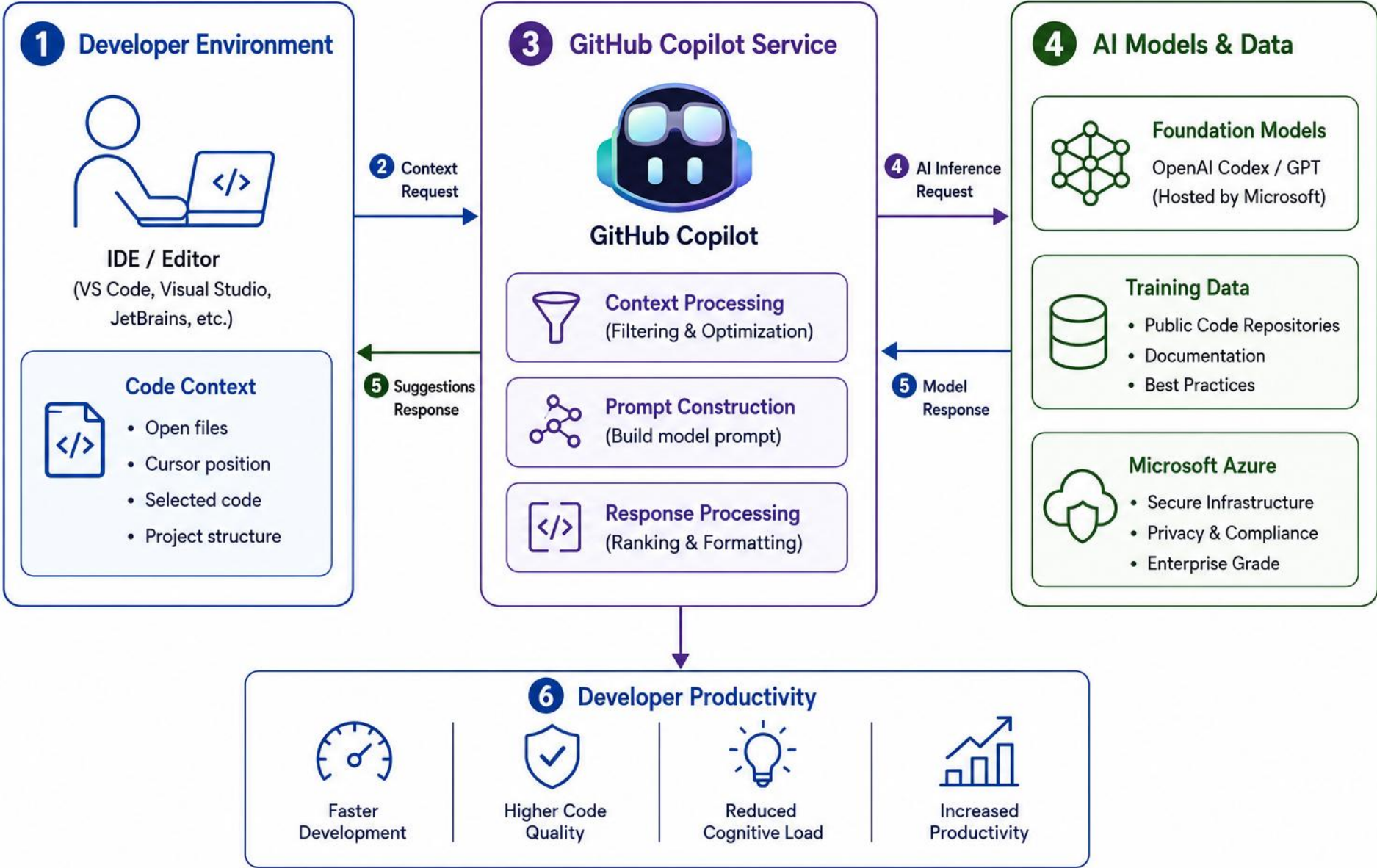
- IDE / Editor — extends your IDE, captures context.
- Context Engine — selects relevant code, filters sensitive data.
- Secure Gateway — encrypts and transmits data.
- AI Service — a selected model generates a response.

SECURITY & PRIVACY

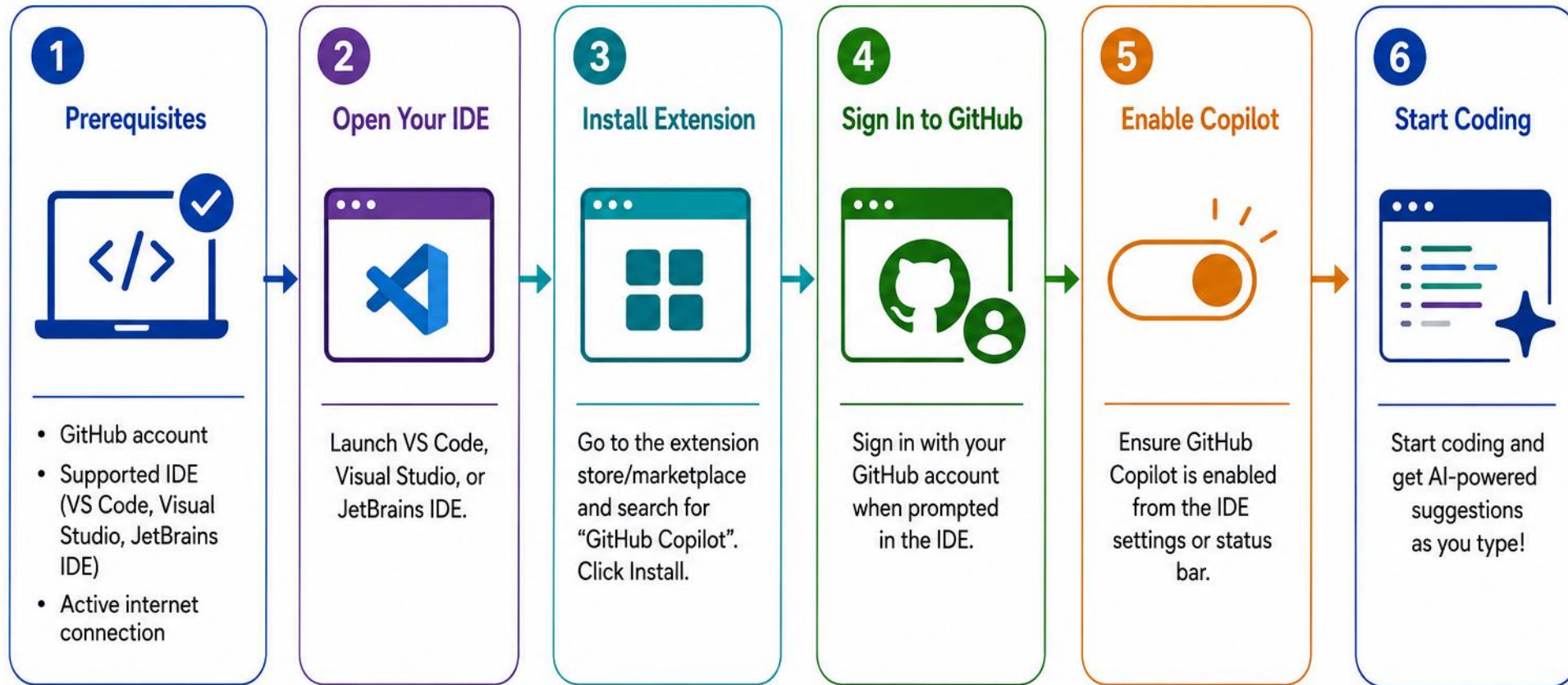
- Data handling depends on your Copilot plan, org policy, settings, and current product terms.
- Follow your organization's Copilot policy at all times.
- Never place secrets, credentials, or regulated data in prompts.

KEY TAKEAWAY Copilot combines large language models, secure cloud infrastructure, and deep IDE integration to deliver real-time assistance.

Copilot Architecture



Installing GitHub Copilot



Note: You need an active GitHub Copilot subscription to use the service.

COPILOT IN INTELLIJ IDEA

Copilot integrates seamlessly with IntelliJ IDEA to help you write, understand, and improve code with AI-powered suggestions.

SETUP IN 3 STEPS

- Go to File > Settings > Plugins > Marketplace, search 'GitHub Copilot', and install.
- Click the Copilot icon in the toolbar and sign in with your GitHub account.
- Authorize Copilot when prompted, then open a file and start typing.

GITHUB COPILOT CHAT

```
> Explain this method.
```

```
This method searches for a user in the  
given list whose ID matches the provided  
id, using Java Streams, and returns an  
Optional<User>. Would you like null checks?
```

TIPS FOR BEST RESULTS



Write Clear Comments

Guide suggestions with intent.



Break Into Small Steps

Complex tasks, one step at a time.



Review Carefully

Before accepting any suggestion.



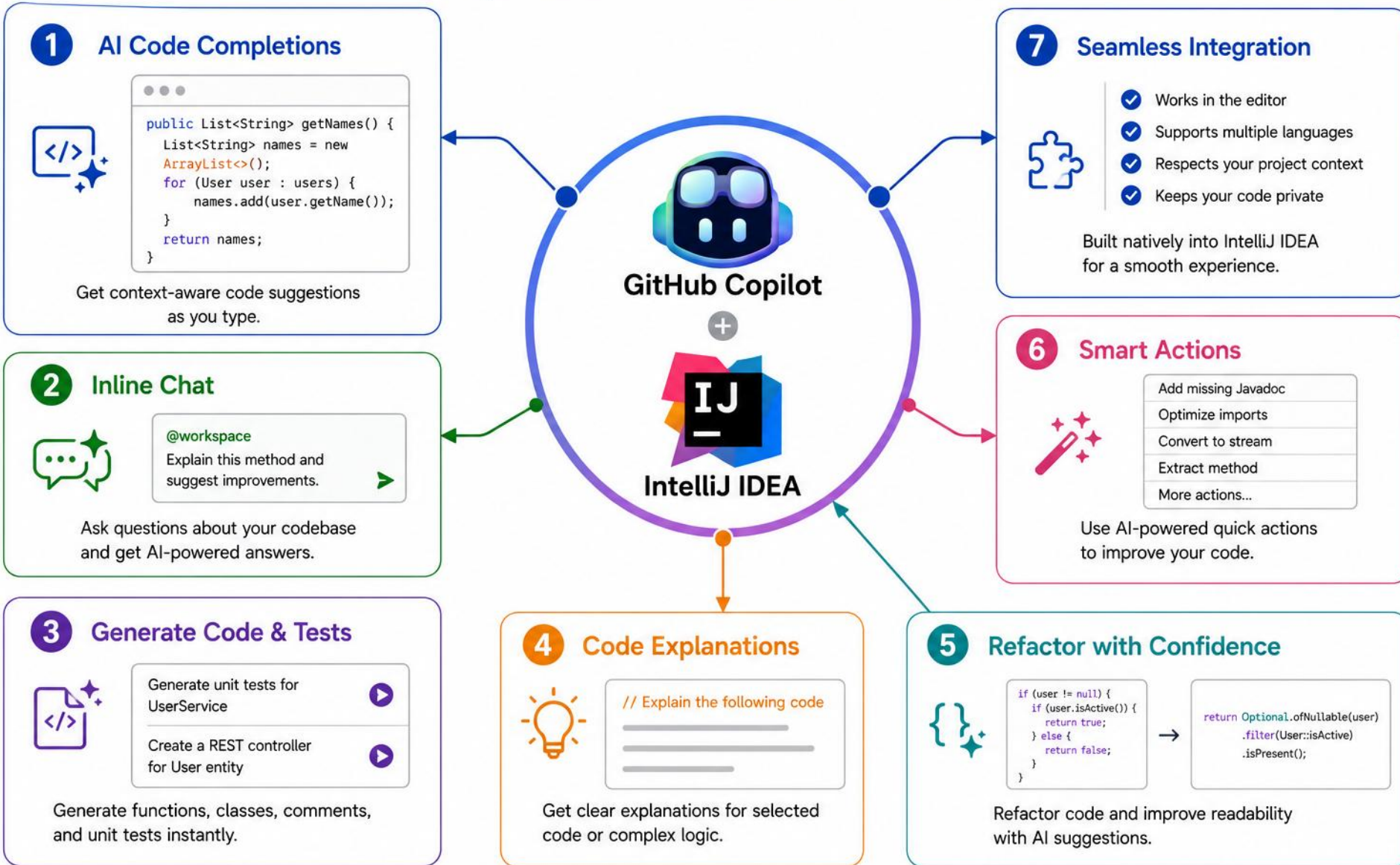
Use Copilot Chat

Learn, refactor, explore alternatives.

- Keyboard shortcuts: Tab to accept a suggestion, Esc to dismiss, Alt+] / Alt+[to cycle through alternative suggestions.

KEY TAKEAWAY Copilot in IntelliJ IDEA acts as your AI pair programmer — understanding context, suggesting code, helping you build faster.

Copilot in IntelliJ IDEA



PROMPT ENGINEERING FUNDAMENTALS

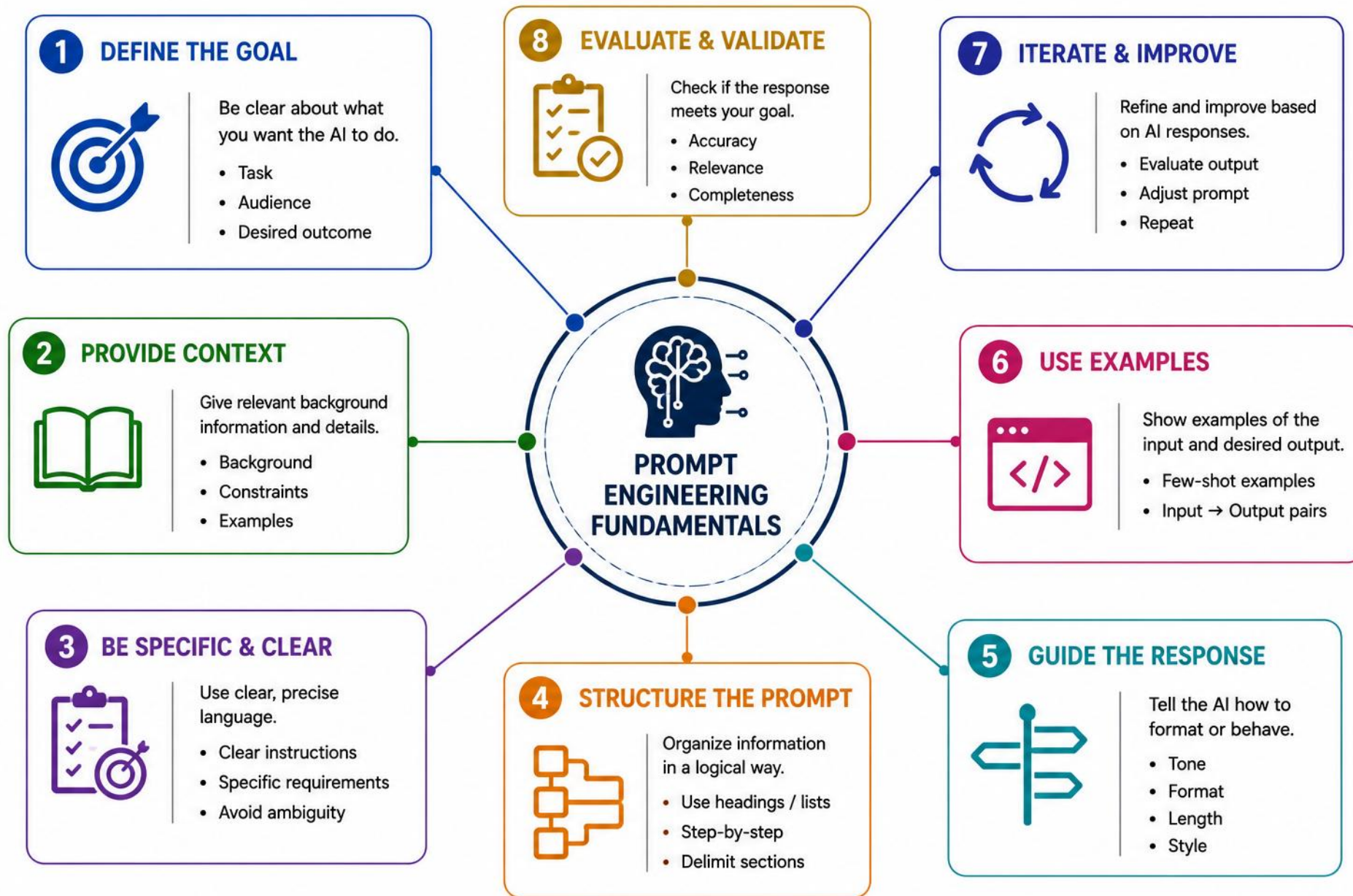
A good prompt helps Copilot understand your intent, use the right context, and generate accurate, high-quality code.



TASK	WEAK PROMPT (VAGUE)	STRONG PROMPT (EFFECTIVE)
Sorting	"Sort a list."	"Sort a List<Employee> by salary, descending, using Comparator."
API Endpoint	"Create an API to get users."	"Create a Spring Boot GET /users/{id} returning JSON, 404 if not found."
Bug Fixing	"Fix this error."	"NullPointerException in UserService line 45 — user can be null; fix safely."

- Prompt improvement workflow: Write Prompt → Get Response → Review Output → Refine Prompt → Better Output (repeat as needed).

KEY TAKEAWAY Great prompts = goal + context + constraints + expected output + edge cases + acceptance criteria.



GENERATING DOMAIN CLASSES AND DTOS

Copilot can scaffold a domain class and its DTO from a clear prompt — you review field types, validation, and where the boundary belongs.

HOW TO GENERATE A CLASS OR DTO

- Write a clear comment describing the class you want to generate.
- Begin typing the class declaration — Copilot suggests the rest.
- Press Tab (or Enter) to accept the suggested code.
- Review the generated code and refine names, fields, or logic.

EXAMPLE PROMPT

```
// Create a Java class Customer with customerId  
// (Long), fullName (String), email (String),  
// phone (String), status (CustomerStatus),  
// createdAt (LocalDateTime). Include getters,  
// setters, constructors, and toString().  
public class Customer {
```

DTO ANNOTATIONS — WHAT GOES WHERE



Bean Validation

@NotNull, @NotBlank, @Size.



Lombok

@Getter, @Builder,
@RequiredArgsConstructor.



Jackson

@JsonInclude, @JsonProperty.



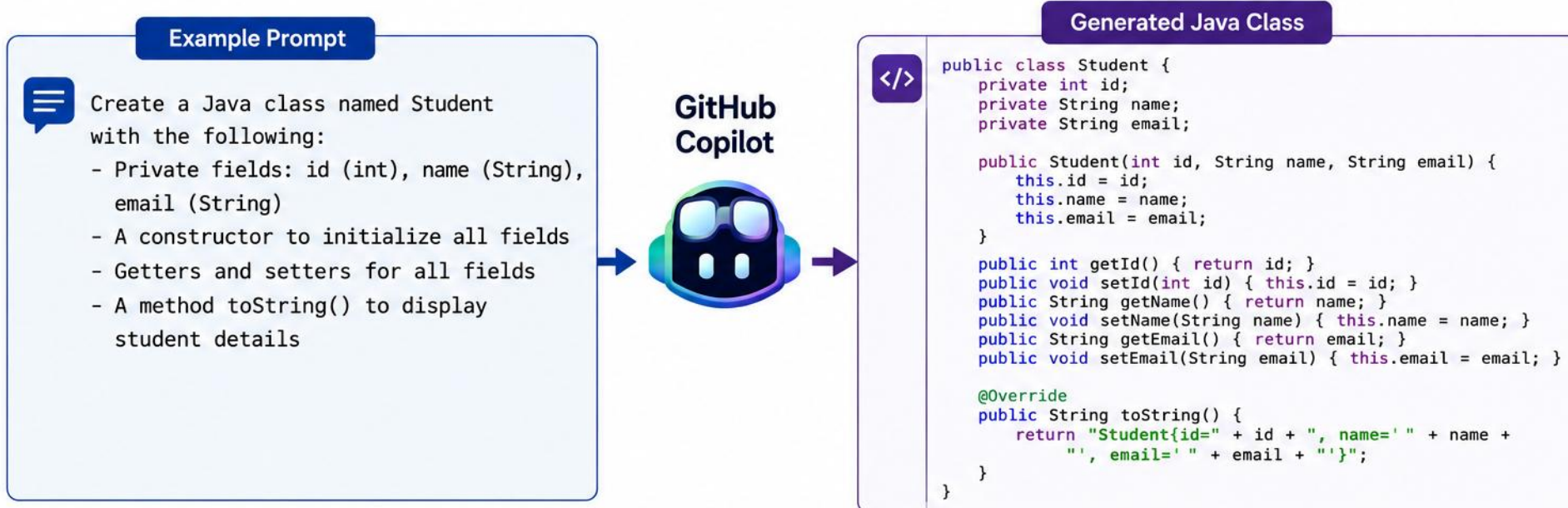
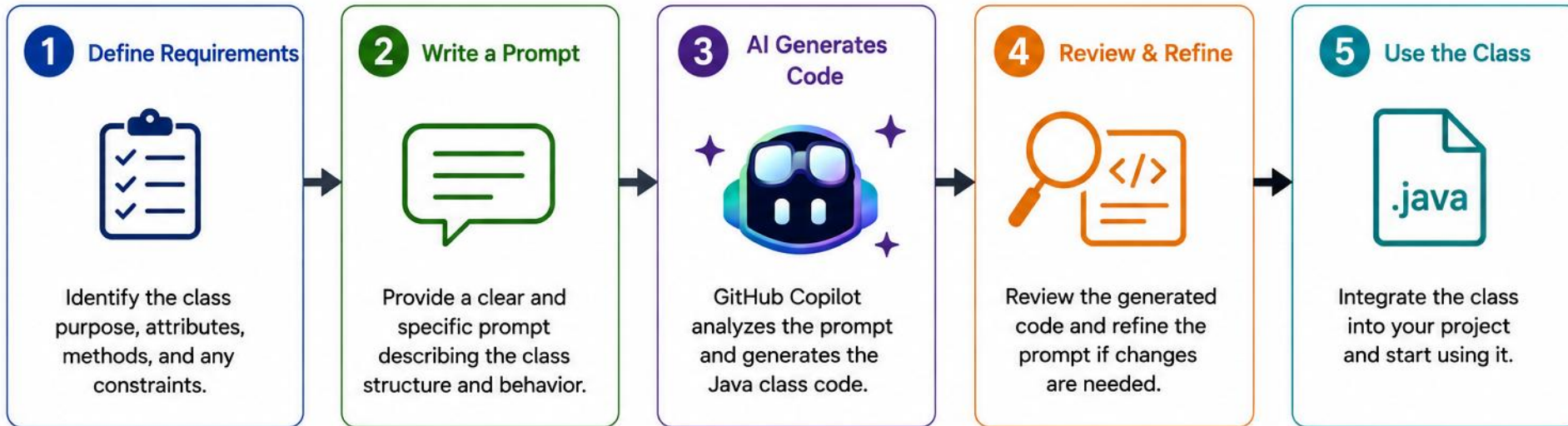
Avoid @Data by Default

Also sets equals/hashCode/toString.

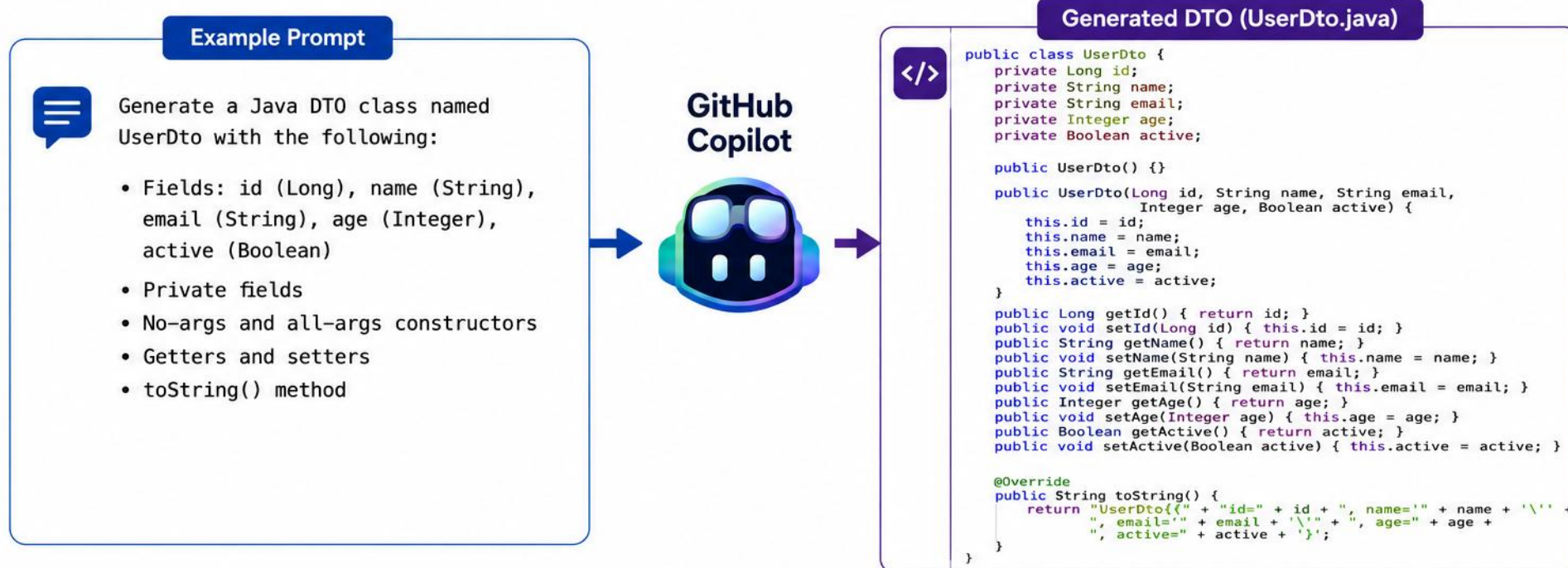
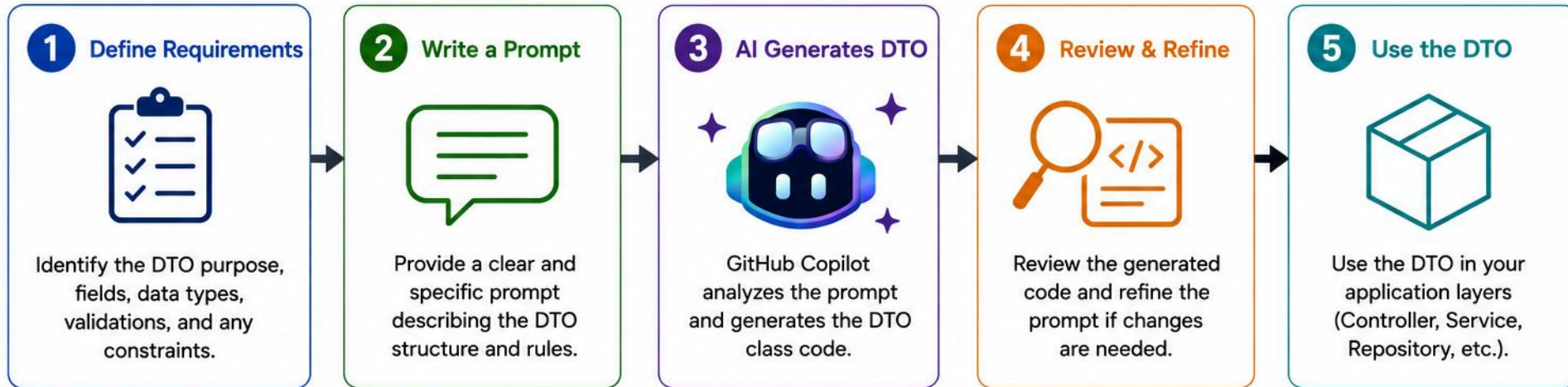
- DTO PROMPT: "Create a DTO for creating a customer with validation annotations for fullName and email." DIFFERENCES TO REVIEW: a DTO carries no business logic — keep it focused on only the fields the API needs, map to/from Customer explicitly, and keep request/response DTOs separate from the entity.

KEY TAKEAWAY Copilot drafts a first pass of a class or DTO — review, test, and align it with your standards before production use.

Generating Java Classes



Generating DTOs



GENERATING SERVICE LOGIC

Service classes hold business logic — validation, search/filter/update rules, and error handling — that Copilot can draft and you refine.

HOW TO GENERATE A SERVICE

- Describe the service's responsibilities, methods, and key dependencies.
- Begin typing the class name or key methods — Copilot suggests the rest.
- Focus on business logic — Copilot fills in method bodies and structure.
- Review generated code and refine error handling and validations.

CUSTOMERSERVICE — PLAIN JAVA



Service Prompt

Add, findById, search, updateStatus.



Business Rules & Errors

Validate input; throw clear exceptions.



Testability

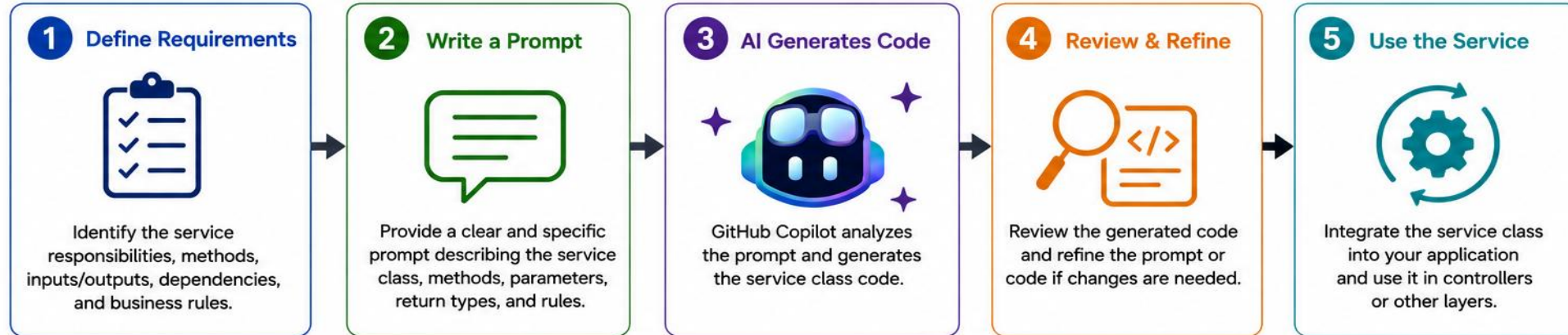
Inject the list; unit test with JUnit.

INDUSTRY PREVIEW — SPRING BOOT ANNOTATIONS

ANNOTATION	PURPOSE
@Service	Marks a business-layer component.
@Repository	Marks a data-access layer component.
@Transactional	Runs the method in a transaction.
@Valid	Validates method parameters.
@RequiredArgsConstructor	Generates constructor (Lombok).

KEY TAKEAWAY Copilot can draft service logic fast — you still own validation, error handling, and tests before it ships.

Generating Service Classes



Example Prompt



Generate a Java Spring service class named UserService with the following:

- Use @Service annotation
- Dependency: UserRepository userRepository
- Methods:
 - UserDto getUserById(Long id)
 - List<UserDto> getAllUsers()
 - UserDto createUser(UserDto userDto)
 - UserDto updateUser(Long id, UserDto userDto)
 - void deleteUser(Long id)
- Use User entity and UserDto
- Add basic business logic and mapping between Entity and DTO

GitHub Copilot



Generated Service Class (UserService.java)

```

</>
@Service
@RequiredArgsConstructor
public class UserService {
    private final UserRepository userRepository;

    public UserDto getUserById(Long id) {
        User user = userRepository.findById(id)
            .orElseThrow(() -> new ResourceNotFoundException("User not found"));
        return mapToDto(user);
    }

    public List<UserDto> getAllUsers() {
        return userRepository.findAll().stream()
            .map(this::mapToDto)
            .toList();
    }

    public UserDto createUser(UserDto userDto) {
        User user = mapToEntity(userDto);
        User saved = userRepository.save(user);
        return mapToDto(saved);
    }

    public UserDto updateUser(Long id, UserDto userDto) {
        User user = userRepository.findById(id)
            .orElseThrow(() -> new ResourceNotFoundException("User not found"));
        user.setName(userDto.getName());
        user.setEmail(userDto.getEmail());
        user.setActive(userDto.getActive());
        return mapToDto(userRepository.save(user));
    }

    public void deleteUser(Long id) {
        userRepository.deleteById(id);
    }

    private UserDto mapToDto(User user) {
        return new UserDto(user.getId(), user.getName(), user.getEmail(), user.getActive());
    }

    private User mapToEntity(UserDto dto) {
        return new User(null, dto.getName(), dto.getEmail(), dto.getActive());
    }
}
  
```

GENERATING DOCUMENTATION

Clear documentation makes your code easy to understand, maintain, and use — Copilot can generate it in seconds.

REVIEW CHECKLIST — BEFORE YOU SHIP DOCUMENTATION

- Does the documentation match the code's actual behavior?
- Are the parameter and return descriptions accurate?
- Are exceptions documented?
- Are setup commands verified?
- Does it avoid stating unsupported guarantees?

WHAT COPILOT CAN GENERATE



JavaDoc

Method and class-level docs.



README

Project overview and setup guides.



API Docs

REST endpoint documentation.



Inline Comments

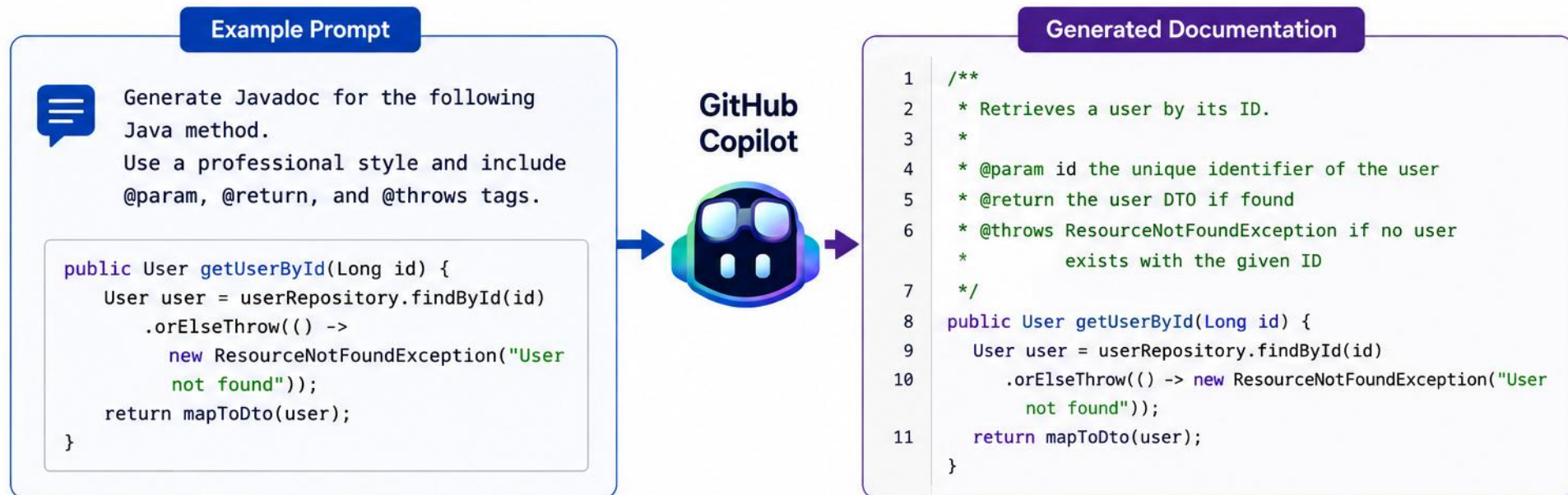
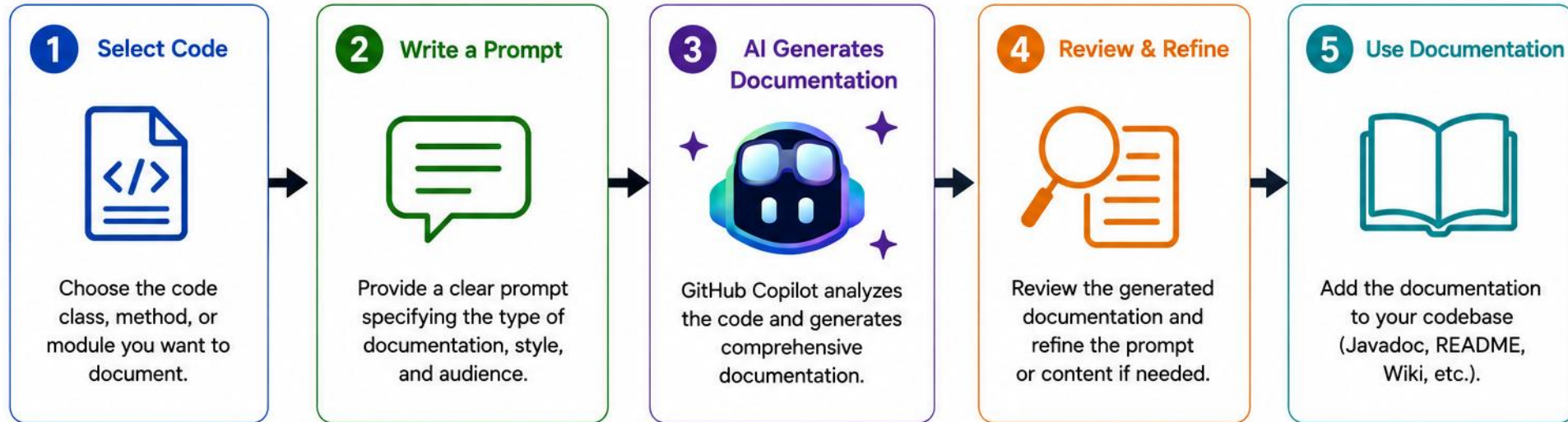
Explain non-obvious logic.

TIPS FOR BETTER DOCUMENTATION

- Be specific in your prompt (e.g. "Generate JavaDoc for this method"). Keep docs in sync with code changes, and follow your project's documentation style and conventions.

KEY TAKEAWAY Good documentation is as important as good code — with Copilot, you can generate high-quality docs quickly.

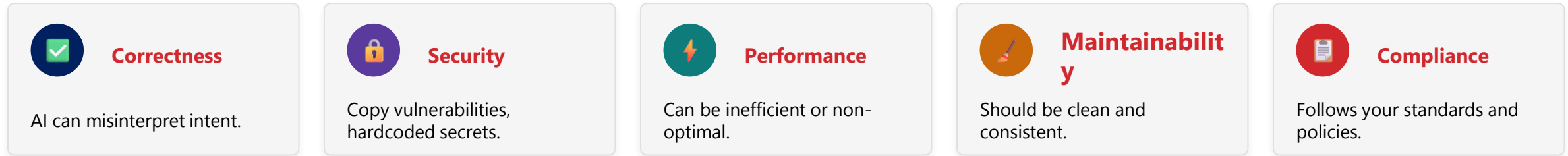
Generating Documentation



REVIEWING AI-GENERATED CODE

AI-generated code can save time, but it's your responsibility to review it for correctness, security, and maintainability. Do: use meaningful names and provide clear, complete prompts. Don't: accept a suggestion blindly or copy code you don't understand.

WHY YOU MUST REVIEW



REVIEW WORKFLOW

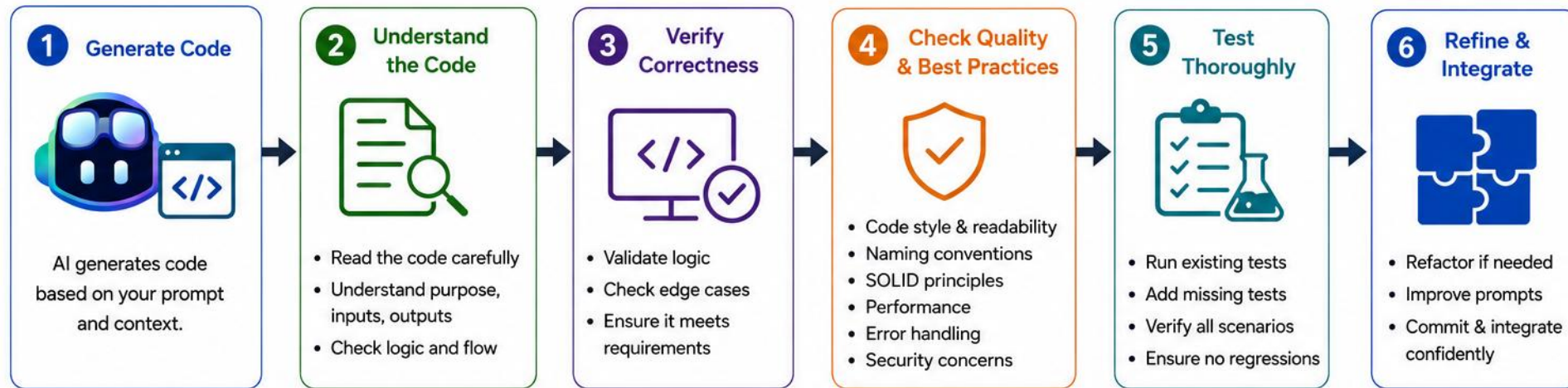


EXAMPLE: PREPAREDSTATEMENT VS STRING CONCATENATION

```
// UNREVIEWED – SQL injection risk
String sql = "SELECT * FROM user WHERE id=" + id;
// REVIEWED & IMPROVED – parameterized query
String sql = "SELECT * FROM users WHERE id = ?";
try (PreparedStatement stmt = connection.prepareStatement(sql)) {
    stmt.setString(1, id);
}
```

KEY TAKEAWAY Never blindly trust AI-generated code. Always understand before you accept — review smart, deploy with confidence.

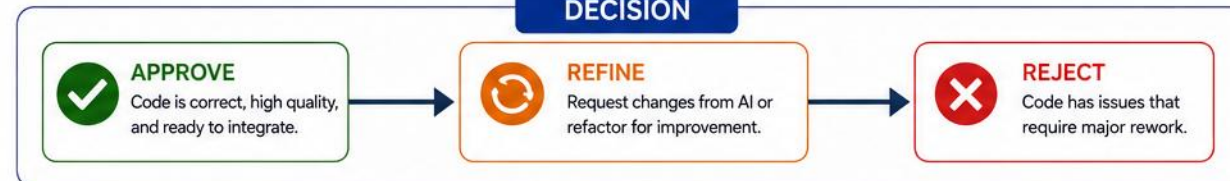
Reviewing AI-Generated Code



Code Review Checklist

 FUNCTIONALITY	 CODE QUALITY	 PERFORMANCE	 SECURITY	 TESTING	 DOCUMENTATION
✓ Does the code meet the requirement? ✓ Are inputs/outputs handled correctly? ✓ Are edge cases considered? ✓ Is the logic correct and complete?	✓ Is the code readable and well-structured? ✓ Are names meaningful and consistent? ✓ Is duplication avoided? ✓ Is the code easy to maintain?	✓ Is the algorithm efficient? ✓ Are loops/queries optimized? ✓ Is resource usage appropriate?	✓ Are inputs validated and sanitized? ✓ Is sensitive data protected? ✓ Are there any security vulnerabilities?	✓ Are unit tests present and relevant? ✓ Do tests cover normal, boundary & error cases? ✓ Do all tests pass? ✓ Any new tests needed?	✓ Is the code documented appropriately? ✓ Are comments clear and helpful? ✓ Is the intent clear to future readers?

DECISION



RISKS OF AI-ASSISTED DEVELOPMENT

AI tools boost productivity but introduce new risks — understanding them helps you use AI responsibly.



Incorrect or Inaccurate Code

- Syntactically correct but logically wrong or incomplete.



Security Vulnerabilities

- Insecure patterns or exposed sensitive data.



License & IP Risks

- May reproduce copyrighted code snippets.



Over-Reliance on AI

- Reduced critical thinking and learning.



Lack of Context Awareness

- Doesn't understand full system or business rules.



Data Privacy Concerns

- Prompts may leak sensitive or proprietary data.



Bias & Ethical Concerns

- May reflect training-data bias, non-inclusive outcomes.



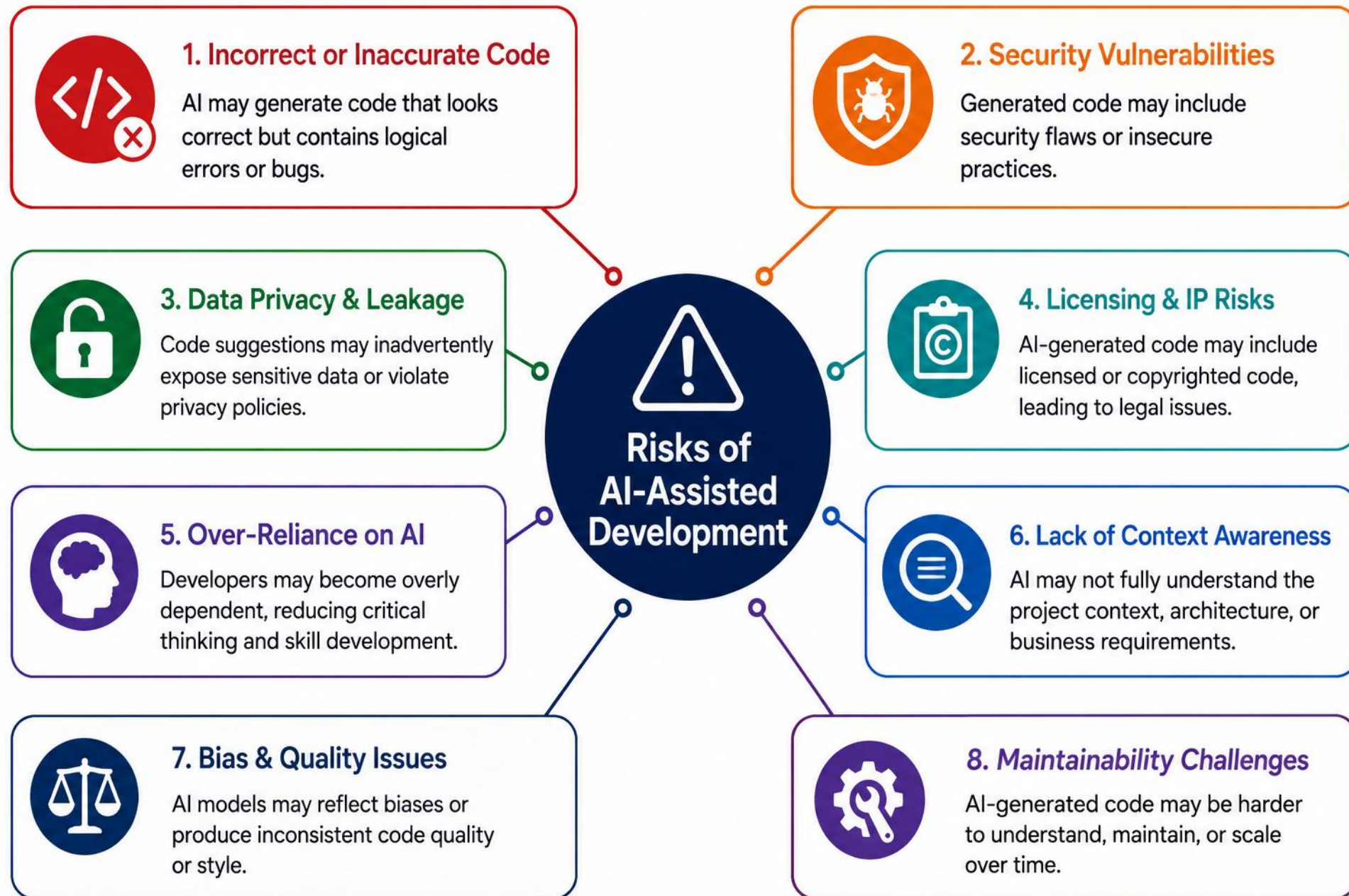
Poor Code Quality

- Can be verbose or misaligned with project standards.

PRACTICAL CONTROLS FOR LICENSE & IP RISK (NOT LEGAL ADVICE)

- Review suggestions for unexpectedly long or distinctive code, and follow your organization's IP policy.
- Use code-referencing/filtering features where available, and validate third-party license compatibility.
- Escalate uncertain cases to legal/compliance.

KEY TAKEAWAY AI is a powerful assistant, not a replacement for engineering judgment. Review, verify, validate — you own the outcome.



ENTERPRISE USE CASES

GitHub Copilot supports Java teams across the development lifecycle — from everyday scaffolding to enterprise-scale delivery.



Refactoring Legacy Java

Modernize old modules and update patterns.



Migration Assistance

Help port code between frameworks or versions.



API & Library Exploration

Discover unfamiliar APIs and libraries faster.



Secure Code Review Support

Detect and fix vulnerable patterns.



Test Automation

Unit, integration, and end-to-end tests.



Explaining Unfamiliar Code

Faster ramp-up on unfamiliar codebases.



Documentation

README, API docs, JavaDoc at scale.



Boilerplate & Scaffolding

Generate classes, DTOs, and config fast.

Faster
Delivery

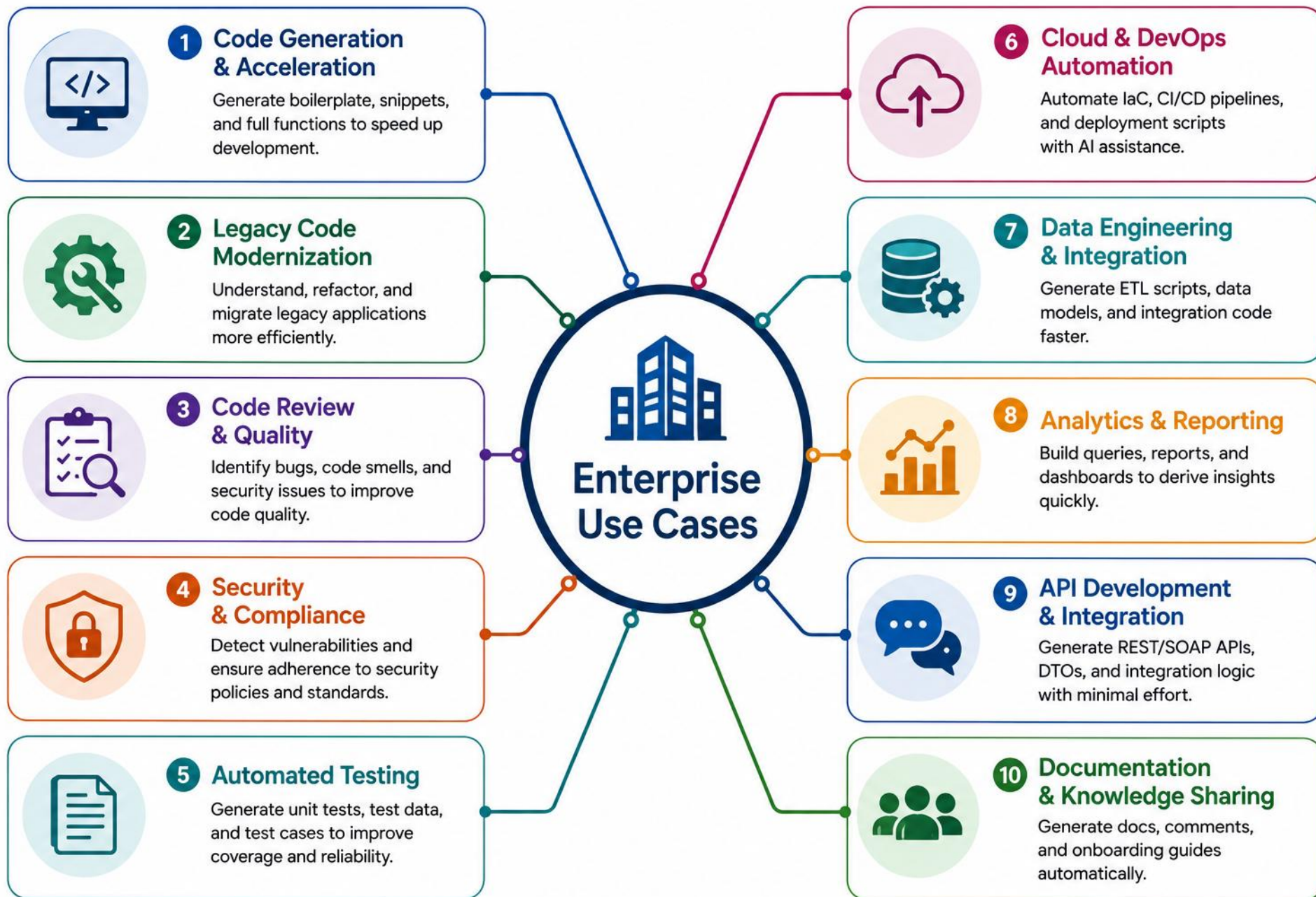
Higher
Quality

Lower
Risk

By Design
Privacy

- Industries: Financial Services, Healthcare, Retail, Manufacturing, Public Sector. Enterprise controls may include organization policies, access management, audit capabilities, content controls, and data-governance options — confirm current capabilities for your GitHub plan.

KEY TAKEAWAY Copilot is more than a coding assistant — it's a force multiplier for engineering teams, used responsibly.



LAB OVERVIEW – AI-ASSISTED CODE GENERATION

Use GitHub Copilot to scaffold the Northstar CRM customer domain — with mandatory human review of every suggestion.

LAB SCENARIO

- Extend the Northstar CRM customer-service Maven project (from Lab 9) with a Customer domain.
- Customer has customerId, fullName, email, phone, status, and createdAt.
- Plain Java 21 and Maven only — no Spring, no JPA, no REST yet.
- Required workflow: Prompt → Review → Refine → Test.

LAB ENVIRONMENT

ITEM	VALUE
Language	Java 21
Framework	None (plain Java + Maven)
Data Store	In-memory List<Customer>
Version Control	Git & GitHub

REQUIRED WORKFLOW



KEY TAKEAWAY Generate, review, refine, and document — use Copilot to accelerate, not to replace your thinking.

LAB TASKS AND DELIVERABLES

Complete the tasks, review and refine the AI-generated code, and deliver a working, reviewed customer domain.

#	TASK	DETAILS	WEIGHT
1	Setup & Sign-In	Sign in to Copilot; copy lab9-crm to lab10-crm; verify Copilot status Ready.	10%
2	Weak vs Strong Prompts	Compare vague vs. specific prompts; log entries lab10-001 / lab10-002.	15%
3	Entity, Enum & Service	Scaffold CustomerStatus, Customer, and CustomerService via Copilot Chat.	25%
4	Main Harness	Prove behavior for CUS-1001 / CUS-1002; mvn clean compile.	15%
5	Human Review & Experiments	Checklist lab10-003; 3+ documented failure experiments.	15%
6	Risk Notes & Docs	ai-review-notes.md lab10-004; security and production answers.	20%

DELIVERABLES

- Customer entity, CustomerStatus enum, and CustomerService compiling clean; a Main harness proving CUS-1001 / CUS-1002; ai-review-notes.md with entries lab10-001–004; no secrets or target/ committed.
- ☐ **Required in ai-review-notes.md:** one accepted suggestion, one modified suggestion, and one rejected suggestion, with the reason for each decision and which tests validated the final code.

KEY TAKEAWAY AI tools can accelerate development, but your judgment, review, and testing deliver quality.

COPILOT DEFINITION OF DONE

Before you call a Copilot-assisted change finished, check it against this practical checklist.

- 1 Prompt includes the goal, context, constraints, and acceptance criteria.
- 2 The suggestion is understood — not just accepted.
- 3 Code follows your project's standards and conventions.
- 4 Tests cover normal cases and edge cases.
- 5 Security and privacy have been reviewed.
- 6 Dependencies and APIs used are verified as real.
- 7 Documentation matches the code's real behavior.
- 8 A human reviewer approves the final change.

KEY TAKEAWAY AI suggests, you decide. Review every suggestion, understand before you accept, and build with quality and standards.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of GitHub Copilot concepts, best practices, and AI-assisted development.

1. What's the primary benefit of GitHub Copilot?

- A. It replaces developers
- B. It automatically tests the application
- C. It writes code faster and reduces busywork**
- D. It deploys the application to production

3. What's a best practice for writing prompts?

- A. Be vague to get creative results
- B. Provide clear context and specific requirements**
- C. Ask Copilot to write the entire app in one prompt
- D. Avoid describing the expected output

2. What should you do right after Copilot generates code?

- A. Review and understand the generated code**
- B. Run it in production immediately
- C. Delete it and start over
- D. Share it on social media

4. What should you always verify in AI-generated code?

- A. Variable and method names
- B. Syntax and code style
- C. Business logic correctness and security
- D. All of the above**

KEY TAKEAWAY Copilot helps developers write code faster — but correctness, security, and style all still need human review.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. What should you do after Copilot generates code?

- A. Accept it as-is without any review
- B. Run it in production to test
- C. Review, understand, test, and refine it**
- D. Share it with your team immediately

7. What's a recommended enterprise Copilot practice?

- A. Share all prompts and code publicly
- B. Follow organization coding standards**
- C. Disable code reviews
- D. Ignore security scanning

ANSWER KEY

- 1-C 2-A 3-B 4-D 5-C 6-B 7-B 8-B

6. Which layer holds business logic in a REST API?

- A. Controller Layer
- B. Service Layer**
- C. Repository Layer
- D. DTO Layer

8. What is the main goal of refining AI-generated code?

- A. To make it longer
- B. To improve readability, performance, and maintainability**
- C. To change variable names only
- D. To avoid writing any code manually

KEY TAKEAWAY Use Copilot intentionally. Review critically. Build securely. Document clearly. Deliver high-quality software.

MODULE 10 COMPLETE

GitHub Copilot Fundamentals for Java Developers

You can now use GitHub Copilot to generate, review, and refine code responsibly — a force multiplier, not a replacement for judgment.